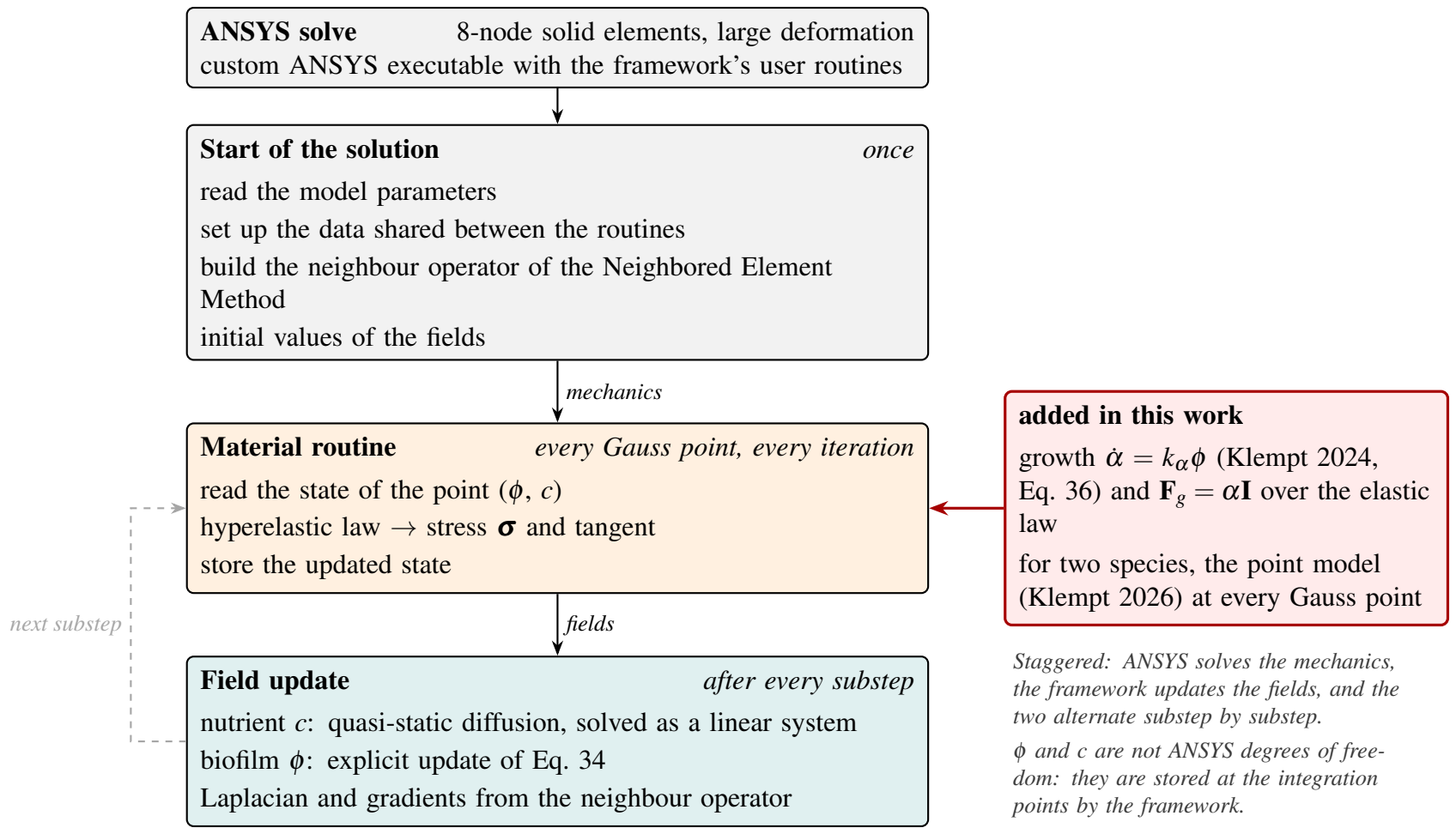

ANSYS solve 8-node solid elements, large deformation
custom ANSYS executable with the framework's user routines
Start of the solution *once*
read the model parameters
set up the data shared between the routines
build the neighbour operator of the Neighbored Element Method
initial values of the fields
mechanics
Material routine *every Gauss point, every iteration*
read the state of the point (ϕ, c)
hyperelastic law $\rightarrow$ stress $\boldsymbol{\sigma}$ and tangent
store the updated state
fields
Field update *after every substep*
nutrient c: quasi-static diffusion, solved as a linear system
biofilm ϕ: explicit update of Eq. 34
Laplacian and gradients from the neighbour operator
next substep
added in this work
growth $\dot{\alpha} = k_\alpha \phi$ (Klempt 2024, Eq. 36) and $\mathbf{F}_g = \alpha \mathbf{I}$ over the elastic law
for two species, the point model (Klempt 2026) at every Gauss point
Staggered: ANSYS solves the mechanics, the framework updates the fields, and the two alternate substep by substep.
ϕ and c are not ANSYS degrees of freedom: they are stored at the integration points by the framework.